

PRACTICAL COURSE WORKBOOK

Vector Databases for AI Apps

Model, query and explain dependable data

LEVEL	Advanced
GUIDED TIME	5 hours across 12 lessons
PROJECT	a working data model with validated queries
SKILLS	Pinecone, Weaviate, pgvector, RAG, Embeddings, Critical evaluation
REVIEWED	2026-08-25

WHAT YOU WILL BE ABLE TO DO

- Use Pinecone appropriately in a realistic, bounded task.
- Use Weaviate appropriately in a realistic, bounded task.
- Use pgvector appropriately in a realistic, bounded task.
- Use RAG appropriately in a realistic, bounded task.

WHO THIS IS FOR

Developers and analysts who need dependable data systems.

BEFORE YOU BEGIN

- Basic programming or spreadsheet experience

Use this guide beside the interactive lessons. Keep inputs, decisions, tests, limitations and the next improvement.

COURSE MAP

From foundation to finished work

Vector Databases for AI Apps is a structured, practical course covering Pinecone, Weaviate, pgvector, RAG, Embeddings. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Data modelling	1. Entities with Pinecone 2. Relationships with Weaviate 3. Constraints with pgvector
02 Querying	1. Selection with RAG 2. Joins with Embeddings 3. Aggregation with Pinecone
03 Reliability	1. Transactions with Weaviate 2. Indexes with pgvector 3. Backups with RAG
04 Production design	1. Security with Embeddings 2. Performance with Pinecone 3. Operations with Weaviate

HOW TO STUDY EACH LESSON

- 1 Read the objective and identify the decision it supports.
- 2 Reproduce the worked example with the stated input.
- 3 Test one normal case and one boundary or failure case.
- 4 Complete the knowledge check without guessing.
- 5 Save a short evidence note: result, limitation and next action.

CAPSTONE PROJECT

Produce a reviewable Vector Databases for AI Apps project using Pinecone, Weaviate, pgvector.

DELIVERABLE	A working Vector Databases for AI Apps artefact plus an evidence-based self-review.
TOOLS	A suitable Pinecone environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using Pinecone.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

Evidence log

INPUT	Original brief, data, requirements or test material.
DECISIONS	Chosen method and one reasonable alternative.
TESTS	Normal, boundary and failure results.
LIMITATION	What the result does not establish.
REVISION	One evidence-led improvement.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

Authoritative references

PostgreSQL Tutorial

PostgreSQL Global Development Group - reviewed 2026-08-21
<https://www.postgresql.org/docs/current/tutorial.html>

SQL Language

PostgreSQL Global Development Group - reviewed 2026-08-21
<https://www.postgresql.org/docs/current/sql.html>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Pinecone scope.